

AY25 PROGRAM ASSESSMENT REPORT

v2026.1

Tracking Data

CIM code (e.g., ME-CM)	MS-BME
Program Name	Master's of Science in Biomedical Engineering
Assessment Year	AY23-25
Program Assessment Coordinator	Kenneth Tichauer
Year of Current Assessment Plan	2025
No Assessment This Year per Plan	☐
LO(s) Assessed	LOs 1-4
Presented at program authority meeting date: <i>Please attach minutes</i>	Report shared with BME faculty by email on October 8 th , 2025. Discussion to be held at following faculty meeting (TBD). Minutes can be appended following meeting.

Methodology

For each LO and artifact assessed, please describe methodology used, and attach (as supplemental files during report submission) or insert here the rubric used. At minimum this should contain:

- a description of how the artifact is concordant with the LO
- a rubric describing the method used to place individual student in at least 3 assessment categories; {**D** Does not meet expectations, **M** Meets expectations *M*, **P** Proficient}

Assessment Administration

- **Learning Objectives 1–4 (LO1–LO4):** Each Fall semester, all MS-BME students are required to complete a program assessment exam in order to receive their course registration PIN. This exam contains questions explicitly mapped to LO1–LO4. The results are used solely for program assessment and do not affect students' grades or degree progress.
- **Learning Objective 5 (LO5):** This LO is assessed during the MS Thesis Defense Exam, in which students demonstrate research proficiency, synthesis of knowledge, and scholarly communication.

Rubric and Scoring Process

- Each exam question was scored independently by **three faculty raters** using the program rubric (1 = Beginning, 2 = Developing, 3 = Proficient, 4 = Exemplary). See attached scoring rubric.
- Scores from the three raters were **averaged** for each student response to ensure reliability and reduce individual grader bias.
- Average scores were then converted into the three institutional categories according to established cutoffs:
 - **Does Not Meet Expectations (D):** < 2.3
 - **Meets Expectations (M):** ≥ 2.3 and < 3.5
 - **Proficient/Excels (P):** ≥ 3.5
- Results were reported at both the **individual question level** and aggregated to the **Learning Objective level**.

Concordance of Artifacts with Learning Objectives

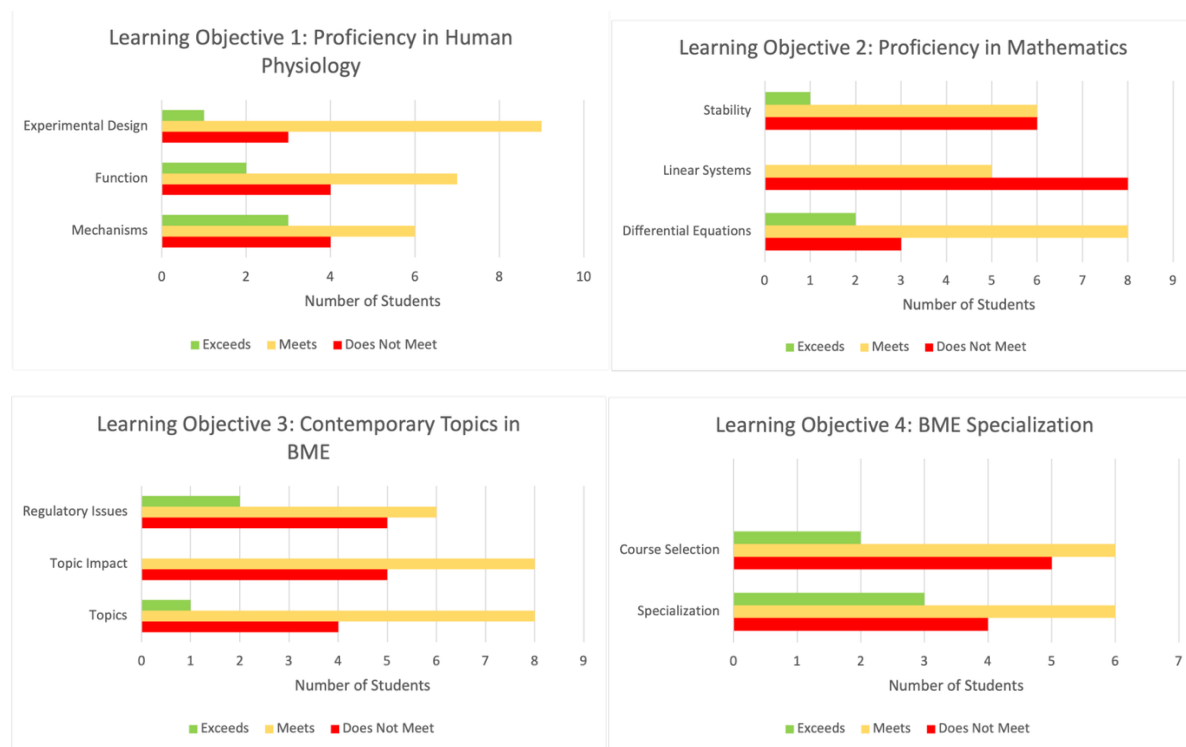
- **LO1 – Be proficient in human physiology:** Exam questions (Q7–Q9) require students to describe organ system mechanisms, explain parameter effects on homeostasis, and design a physiology experiment. These tasks directly measure physiological understanding and application.
- **LO2 – Be proficient in advanced mathematics:** Exam questions (Q10–Q12) assess techniques for solving ODEs, application of least squares regression, and interpretation of eigenvalues, providing evidence of quantitative proficiency.
- **LO3 – Breadth of understanding of contemporary BME topics:** Exam questions (Q13–Q15) ask students to identify recent advancements, assess the impact of emerging technologies, and evaluate ethical/social/regulatory issues, reflecting expectations for broad field awareness.
- **LO4 – Mastery of a specific BME topic:** Exam questions (Q16–Q17) require description of an elective specialization and integration of electives into a coherent professional path, aligning with the expectation for focused expertise.
- **LO5 – Demonstrated research proficiency:** The MS Thesis Defense examination provides direct evidence of research design, execution, interpretation, and communication.

Results

Artifact samples

Please see attached form: **AY25 MS-BME Prog Assessment Report Form_all_artifacts_for_LO1-4.pdf**

Data table



Description of findings

When the results were presented to the BME faculty, discussion centered on the encouraging performance in **physiology (LO1)**, where the majority of students were rated as *Meets Expectations* or *Proficient/Excels* across all three physiology questions. Only a small number of students fell below expectations, suggesting that the physiology core is being delivered effectively.

In contrast, **mathematics (LO2)** emerged as a clear area of concern. While most students were able to meet expectations on basic ODE solution techniques (Q1), performance was notably weaker on questions related to linear systems and stability analysis. For example, 8 of 13 students scored *Does Not Meet Expectations* on the linear systems question, with only 5 meeting expectations and none excelling. This result indicates a significant gap in student mastery of advanced mathematical tools, which are critical for BME problem-solving. Faculty discussed possible contributing factors, such as uneven prior preparation in mathematics

among incoming MS students and the placement of quantitative material within the curriculum. Follow-up actions under consideration include strengthened prerequisites, review modules, or integration of more applied math assignments into required courses.

For **contemporary BME topics (LO3)**, results showed that students can generally identify recent advancements and describe ethical/regulatory issues, but many did so only at a superficial level. The majority met expectations, though relatively few excelled. Faculty agreed this reflects adequate exposure to contemporary developments but also a need for greater emphasis on critical evaluation and synthesis of these topics, perhaps through journal club-style assignments or structured seminar reflections.

For **specialization (LO4)**, most students were able to articulate their elective area and relate coursework to a professional trajectory, with a mix of meets and exceeds. A small subset did not demonstrate sufficient coherence in their elective pathway planning. Faculty viewed this as an opportunity to improve academic advising and to provide clearer guidance on how electives can be integrated into a coherent specialization.

Summary:

- **Strengths:** Physiology (LO1) shows solid performance overall; students demonstrate adequate awareness of contemporary BME topics (LO3); most students can describe and integrate their specialization (LO4).
- **Concerns:** Quantitative skills (LO2) are the most significant weakness, especially in linear systems and stability analysis. Contemporary topics (LO3) and specialization (LO4) are adequate but show limited excellence, suggesting room for growth in higher-order critical thinking and integration.
- **Next Steps:** Program faculty will consider curricular adjustments to reinforce quantitative preparation (e.g., prerequisites, supplemental instruction, or course design changes) and will explore opportunities to strengthen student engagement with contemporary BME literature and elective pathway planning.

Continuous Improvement, CI, plans

The purpose of assessment is to improve the quality of our educational programs. The most important part of the assessment cycle is how we use the data to improve our programs. Examine the data provided above describing student achievement, consider the root cause of any deficits, and if areas for possible improvement emerge, describe plans to achieve this improvement.

Prior CI plan impact

Also, please refer to the previous assessment report for the LOs assessed. Were any CI plans recommended? Were they implemented? Did the plans work (were improvements seen in this cycle)?

☒no previous CI plans available

This cycle's CI plan

Describe any CI plans developed from this cycle's data, with

- Specific curricular or instructional changes in response to this assessment data,
- A timeline for implementation,
- As well a description of who will participate in and manage these plans

☐After consideration and presentation to the program authority, no CI plans are recommended.

Based on the AY25 results, the program faculty identified **advanced mathematics (LO2)** as the most significant area of concern. Student performance on linear systems and stability analysis was especially weak, with more than half of students not meeting expectations. Faculty discussed potential root causes, including variability in students' prior quantitative preparation and insufficient reinforcement of advanced mathematical tools in the MS curriculum.

To address this, the following Continuous Improvement plan was developed:

1. Curricular Adjustments:

- Incorporate additional applied examples of linear systems and stability analysis into the required mathematics/quantitative methods course (BME 522 or equivalent).
- Consider develop a supplemental "Math for Biomedical Engineers" online review module, to be made available to incoming MS students prior to their first semester.
- Encourage faculty teaching advanced BME electives to include short, embedded reviews of key math tools where relevant (e.g., stability concepts in imaging physics, eigenvalue analysis in modeling courses). Tracking plan is still under development.

2. Timeline for Implementation:

- Online review module to be developed by Fall 2026, piloted Spring 2026.
- Curricular adjustments to BME 522 to be implemented beginning Fall 2026.
- Review of impact to be conducted in the Fall 2027 assessment exam cycle.

3. Responsible Parties:

- Program Assessment Coordinator (Ken Tichauer) to coordinate development and monitor outcomes.
- BME Graduate Curriculum Committee to oversee curricular changes.
- Course instructors (BME 522 and relevant electives) to implement applied math reinforcement.

Other Areas Noted:

- For **LO3 (Contemporary BME topics)** and **LO4 (Specialization)**, faculty noted that performance was generally acceptable but with limited evidence of students excelling. To address this, faculty agreed to:
 - Pilot a **journal club / seminar reflection assignment** in Fall 2026, requiring MS students to critically evaluate emerging BME research and discuss ethical/social implications.
 - Strengthen **advising materials** to better guide students in selecting coherent elective pathways that align with their professional goals.

Summary of CI Actions:

- **Major focus:** Reinforce quantitative skills (LO2) through pre-enrollment preparation and embedded curricular adjustments.
- **Secondary focus:** Enhance higher-order critical thinking in contemporary BME topics (LO3) and coherence in specialization planning (LO4).
- **Timeline:** Implementation in AY26, impact to be reassessed in AY27.
- **Responsibility:** Program Assessment Coordinator, Graduate Curriculum Committee, and course instructors.